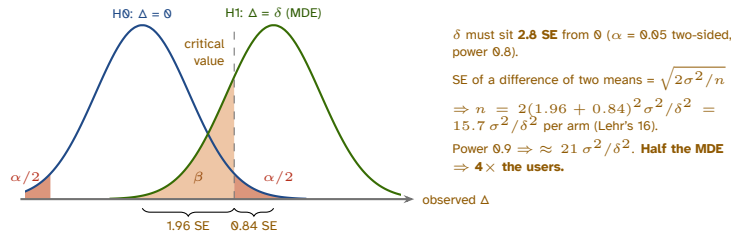
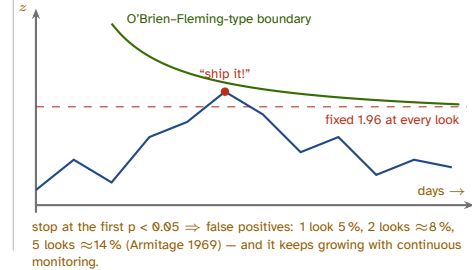


In one line: An A/B test is a **hypothesis test on randomised groups**. You fix α (false-positive rate), **power** $1 - \beta$ (chance to detect the **MDE** if it is real) and the MDE **before** starting — together they fix the **sample size**. Then you prove the experiment is **valid** (SRM, exposure) **before** reading the effect, read it **once** at the planned horizon (unless the test was built for peeking), and report the effect with its confidence interval, not a bare p-value.

Why $n \approx 16 \sigma^2 / \delta^2$: the two curves of the observed difference



Peeking: a null test crosses 1.96 by chance



The vocabulary, exactly

- p-value** = $P(\text{a difference at least this extreme} \mid H_0 \text{ true})$. Not $P(H_0 \mid \text{data})$, not “95% sure it works”.
- α (0.05) = false-positive rate you accept when there is no effect (Type I). β = chance of missing a real effect of size MDE (Type II); **power** = $1 - \beta$, usually 0.8.
- MDE** = smallest effect worth shipping *and* the one the test is sized for. Say whether it is **absolute** (+0.5 pp) or **relative** (+5%).
- 95% CI** excludes 0 $\Leftrightarrow p < 0.05$ (two-sided). Report $\Delta \pm \text{CI}$. Not significant $\neq$ “no effect” — only “smaller than we could see”.

Sample size & duration — worked

- Conversion: $\sigma^2 = p(1 - p)$. Baseline 10%, MDE +0.5 pp: $16 \cdot 0.09 / 0.005^2 = \mathbf{57\,600}$ per arm (exact: $\approx 56\,500$).
- Duration = $n \div$ exposed users per arm per day, rounded up to **whole weeks** (weekday / weekend behaviour differs).
- Metric per session but randomised per user? Sessions of one user are correlated $\rightarrow$ the naive variance is too small; use the **delta method** or bootstrap by user.

```
from scipy.stats import norm, chisquare
def n_per_arm(p, mde, alpha=0.05, power=0.8):
    z = norm.ppf(1 - alpha / 2) + norm.ppf(power)
    return 2 * z**2 * p * (1 - p) / mde**2
n_per_arm(0.10, 0.005) # 56 512 per arm
chisquare([50_800, 49_200]) # stat 25.6, p ~ 4e-7: SRM
```

Peeking & sequential testing

- Fixed horizon**: pick n , look once. Simple, and the default.
- Group sequential** (alpha spending, O'Brien-Fleming-type): planned looks, very strict early, near 1.96 at the end.
- Always-valid p-values / MSPRT**: built to be monitored continuously, at the cost of power. Stopping for **harm** (guardrail breach) at any time is fine.

Sample-ratio mismatch (SRM)

- χ^2 goodness of fit of the *user counts* against the designed split (1 df for 2 arms): $\chi^2 = \sum (O - E)^2 / E$. 50/50 over 100 000: 50 800 vs 49 200 $\rightarrow \chi^2 = 25.6$, $p \approx 4 \cdot 10^{-7}$.
- Microsoft's platform flags **p < 0.0005** and hides results until it passes (Fabijan et al., KDD 2019). An SRM **voids** the result — find the bug, don't reweight.
- Causes: bucketing bug, one arm crashing (telemetry lost), bot filtering, a trigger the treatment itself changes, late exposure logging.

Effects that fool you

- Novelty**: users poke the new thing because it is new — the lift decays. **Primacy** / change aversion: habit makes the new UI lose at first — the lift grows. Detect: effect by day and by first-exposure cohort; new vs returning users; run ≥ 2 weeks.
- Twyman's law**: a result that looks too good is usually a bug — check before celebrating.
- Simpson's paradox**: pooling days with different splits (1% $\rightarrow$ 50% ramp) can reverse the sign — analyse per stage or keep the split fixed.

Guardrails & multiple comparisons

- One pre-registered **primary metric** (the decision), secondary metrics to explain it, **guardrails** that must not regress: crash-free users, hang rate, latency, revenue, uninstalls; trust guardrails: SRM, exposure rate.
- 20 metrics, no real effect, $\alpha = 0.05$: $P(\text{at least one "win"}) = 1 - 0.95^{20} \approx \mathbf{64\%}$. Fixes: Bonferroni α/m (family-wise), Benjamini-Hochberg (false discovery rate). Segment slicing too.

CUPED — variance reduction

- $Y' = Y - \theta(X - E[X])$, X = the same metric in the **pre-experiment** period, $\theta = \text{cov}(X, Y) / \text{var}(X)$.
- Unbiased (X is fixed before assignment); variance $\times (1 - \rho^2)$. $\rho = 0.7 \rightarrow 0.51 \rightarrow$ about **half the users** for the same power. Bing measured $\approx 50\%$ (Deng et al., WSDM 2013). Brand-new users have no $X \rightarrow$ little gain for onboarding tests.

Mobile-specific pitfalls

- Version adoption lag**: only the new build can expose; early updaters may not look like everyone.
- Assignment unit**: install id re-randomises on reinstall / second device; logging in mid-test swaps the id $\rightarrow$ one person sees both arms. Post-login features: user id.
- Offline exposure logging**: events queue and arrive late or never (uninstall, crash). A crashier arm loses its telemetry $\rightarrow$ SRM and a flattering metric. Set a late-arrival cutoff.
- Cached flags**: a new assignment applies at the next launch; users who never relaunch are never treated $\rightarrow$ log exposure when the variant **renders**, analyse exposed users only.

Traps

- ! “ $p = 0.03$ means 97% chance B is better” — no: it is $P(\text{data} \mid \text{no effect})$.
- ! Extending a test “until it's significant” (peeking); reading the metric before the SRM check.

Remember $16\sigma^2 / \delta^2$ per arm • SRM first • look once or go sequential • one primary + guardrails • CUPED • exposure on render.